

PVC-Injectors 0.5 - 4.0 kg/h

Installation and operating instructions



Original installation and operating instructions.

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**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Safety information

Non-observance of the safety instructions may have dangerous consequences for the staff, the environment and the product and may result in the loss of any claims for damages.

Should you require further information, please contact Grundfos.

1.1 Target group

These installation and operating instructions contain general instructions that must be observed during installation, startup, operation and maintenance of the product. The responsible staff must read these installation and operating instructions prior to any work at the product. These installation and operating instructions must be available at the installation location at all times.

1.1.1 Qualification and training of staff

The staff responsible for the installation, startup, operation and maintenance must be appropriately qualified for these tasks. Areas of responsibility, levels of authority and the supervision of the staff must be precisely defined by the operating company. If necessary, the staff must be trained appropriately.

1.1.2 Obligations of the operating company

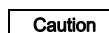
- Observe the local safety regulations.
- Instruct the operating staff.
- Provide the stipulated safety equipment and personal protective equipment.
- Arrange regular maintenance.

1.1.3 Obligations of the user

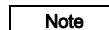
- Read this manual thoroughly before operating the product.
- Observe the recognised health and safety regulations as well as the accident prevention regulations.
- Wear appropriate protective equipment in accordance with national health and safety regulations when working at the system and handling chemicals.

1.2 Symbols used in this document**Warning**

If these safety instructions are not observed, it may result in personal injury.

**Caution**

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

**Note**

Notes or instructions that make the job easier and ensure safe operation.

1.3 Safety of the system in the event of product failure

If the product fails, the safety of the overall system must be ensured. Use appropriate monitoring and control functions.

Make sure, that leaking chemicals do not cause personal injury or damage to property.

**Caution**

The installation of leak monitoring solutions and drip trays is recommended.

1.4 Working with chemicals

Warning

Danger of personal injury due to contact with chemicals!



Wear the stipulated personal protective equipment (protective clothing, goggles, respirator etc.) when handling chemicals!

Observe the chemical manufacturer's safety data sheets (MSDS) and safety instructions of the used chemicals!

Make sure, that parts in contact with the chemicals are resistant to the chemicals under the specific operating conditions!

Caution

Should you have questions regarding the material resistance of the product for specific chemicals, please contact Grundfos.

2. Product description

2.1 Applications

2.1.1 Intended use

Grundfos PVC injectors are intended for use within Grundfos gas dosing systems for water treatment, strictly complying with the instructions in this manual. The injector provides the necessary operating vacuum in the gas dosing system. The vacuum sucks the gaseous dosing medium into the injector where it is dissolved in the process water.

Media

The product is intended for use with one of the following media:

- Cl₂
- NH₃
- CO₂
- SO₂

The allowed medium is specified on the nameplate. The media must have the following characteristics:

Aggregate state	Chemical characteristics
• gaseous	• non-combustible • non-abrasive • non-crystallising

Observe the freezing and boiling point of the used medium at operating pressure. Observe the temperatures specified in section 3. *Technical data*.

2.1.2 Improper use



Warning

Improper use, misuse or misapplication can lead to personal injury and damage to the equipment!

The operational safety of the product is only guaranteed, if it is used in accordance with section 2.1.1 *Intended use*.

- The product must not be used for media that are not specified on the nameplate.
- The product must not be used if it is damaged.
- The product must not be used after improper repair.
- The product must not be used after unauthorized modification.

2.2 Nameplate



Fig. 1 Nameplate

Pos.	Description
1	Product number
2	Serial number
3	Max. suction power
4	Counterpressure range
5	Motive water connection (inlet/outlet)
6	Gas connection
7	Code for year and week of production
8	Country of origin
9	Type designation
10	Allowed medium

TM06 1273 1115

3. Technical data

Caution

The values stated in the technical data must be adhered to.

3.1 General data

Dosing media	Cl ₂ / NH ₃ / SO ₂ / CO ₂	
Max. motive water pressure	[bar]	16
Max. counterpressure	[bar]	7 / 10
Permissible motive water temperature	[°C]	+5 to +30
Permissible ambient temperatures (min. / max.)		
Transport	[°C]	-5 / +50
Storage	[°C]	-5 / +50
Operation	[°C]	+5 / +40

3.2 Connection data

3.2.1 Motive water inlet and gas solution outlet connection

The inlet and outlet connections have always the same size. The following options are available:

- DN 20
- 3/4" NPT
- 1" NPT

3.2.2 Gas connection

The following options are available:

- For PE hose 8x11
- For PE hose 10x14
- For PE hose 12x16
- For PVC pipe DN 10
- For PVC pipe DN 15

3.3 Injectors for Vaccuperm systems WITH differential pressure regulator (VGA-113, VGA-117, VGS-141)

3.3.1 Suction power for counterpressure range 0 to 7 bar

Injector variant	Max. suction power [g/h]			
	Cl ₂	NH ₃	SO ₂	CO ₂
up to 500 g/h	500	500	500	210
up to 1000 g/h	1000	1000	1000	350
up to 2000 g/h	2000	1500	2000	550
up to 4000 g/h	4000	2000	4000	800

3.3.2 Suction power for counterpressure range 7 to 10 bar

Injector variant	Max. suction power [g/h]			
	Cl ₂	NH ₃	SO ₂	CO ₂
up to 500 g/h	500	-	-	-
up to 1000 g/h	1000	-	-	-
up to 2000 g/h	2000	-	-	-

3.4 Injectors for Vaccuperm systems WITHOUT differential pressure regulator (VGB-103, VGA-113, VGA-117)

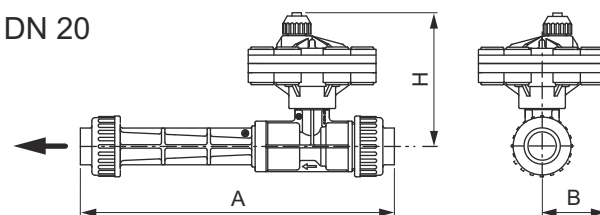
3.4.1 Suction power for counterpressure range 0 to 7 bar

Injector variant	Max. suction power [g/h]			
	Cl ₂	NH ₃	SO ₂	CO ₂
up to 500 g/h	500	500	500	210
up to 1000 g/h	1000	1000	1000	350
up to 2000 g/h	2000	1500	2000	550

3.5 Dimensions

3.5.1 Injectors for counterpressure range 0 to 7 bar

DN 20



NPT

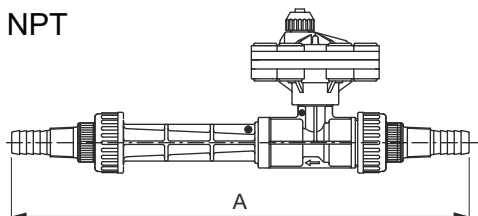
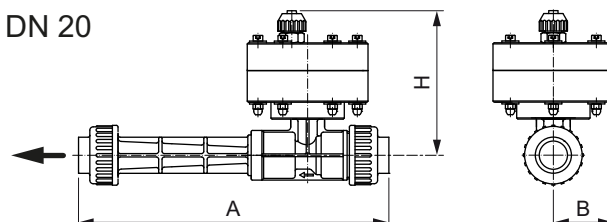


Fig. 2 Example with gas connection for PE hose 8x11

Dimension [mm]		A			B	H
		Motive water connection				
		DN 20	3/4" NPT	1" NPT		
Gas connection	For PE hose 8x11	275	401	429	56	117
	For PE hose 10x14	275	401	429	56	131
	For PE hose 12x16	275	401	429	56	136
	For PVC pipe DN 10	275	401	429	56	145
	For PVC pipe DN 15	275	401	429	56	145

3.5.2 Injectors for counterpressure range 7 to 10 bar

DN 20



NPT

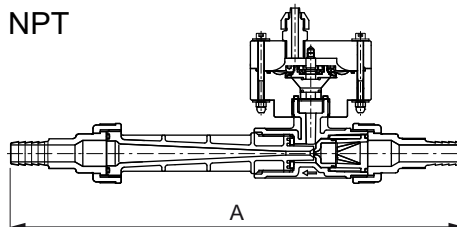


Fig. 3 Injectors with gas connection for PE hose 8x11

Dimension [mm]	A			B	H
	Motive water connection				
	DN 20	3/4" NPT	1" NPT		
Gas connection: For PE hose 8x11	275	401	429	54	128

4. Construction and function

4.1 Function

Injectors operate according to the principle of water jet pumps. The injector body comprises a nozzle with a successional diffuser. Between the nozzle and the diffuser, there is a narrow annular gap, where gas is sucked out of the gas supply line through the injector head. The gas passes the diffuser and is mixed with the motive water in the downstream mixing section. A diaphragm non-return valve prevents the ingress of water into the gas supply line.

4.2 Components

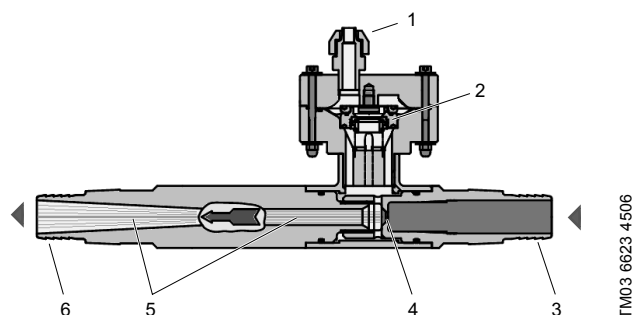


Fig. 4 Injector components

Pos.	Description
1	Gas connection for gas supply
2	Body with diaphragm non-return valve: The diaphragm non-return valve prevents water from entering gas-conveying parts of the device.
3	Motive water connection - inlet
4	Nozzle: Accelerates the motive water flow and creates a low pressure.
5	Diffuser with mixing section: Mixes motive water with gas.
6	Gas solution line connection = motive water outlet

5. Installation

5.1 Safety instructions



Warning

The installation must be carried out by authorised and qualified staff.

Wear the stipulated personal protective equipment (protective clothing, gloves, safety boots)!

5.2 Transport and storage

Caution Handle with care, do not throw!

Store in a dry and cool place.

5.3 Unpacking

Observe when unpacking:

- No humidity should get into gas-leading parts.
- No foreign matter should get into gas-leading parts.
- Mount as soon as possible after unpacking.
- Use appropriate lifting devices to handle heavy parts.

5.4 Installation location

- Choose a mounting place free of vibrations.
- The installation location must be protected from direct sunlight and rainfall.
- Observe the permissible ambient temperature!

5.5 Installation requirements

- Keep the line from the gas solution outlet of the injector to the injection unit as short as possible.
- Observe the prescribed line cross-sections to avoid a drop in output. See www.grundfos.com.

5.5.1 Motive water requirements

- The motive water must be good-quality fresh water.
- The motive water must be free from impurities and abrasive materials. Fit a dirt trap at the beginning of the motive water line, if necessary.
- The motive water pressure at the inlet of the injector as well as the counterpressure at the gas solution outlet must be sufficient to achieve the required dosing rate. See section 11. *Dimensioning tables*. Fit a pressure booster pump into the motive water line, if necessary.
- Make sure, that the permissible motive water pressure as well as the permissible counterpressure cannot be exceeded.
- The motive water temperature must be within the range given in section 3.1 *General data*.

5.5.2 Fitting the injector

This section refers to fig. 4.

All connections must be mounted free from mechanical stress.

Caution

Observe the flow direction (arrow sticker).

1. Fit the injector to the wall or another suitable location using the fixing material supplied.
2. Connect the motive water line at connection (3).
3. Connect the gas solution line at connection (6).
4. Connect the gas supply at connection (1).

6. Commissioning

6.1 Safety instructions

Warning

The device must be commissioned only by qualified staff.

Observe the material safety data sheets (MSDS) of the handled chemicals!

Wear protective clothing and a gas mask when carrying out work at the system!

All gas container valves must be closed.

Make sure, that no residual gas is in the system.

Make sure, that a sufficient amount of neutraliser (e.g. sodium thiosulphate for chlorine) is available on site.

For commissioning, all components of the entire system must be ready for operation.

Observe the manuals of the other components.

Open and close isolating valves in motive water line and gas solution line slowly to avoid hydraulic shocks.



Caution

6.2 Checks before commissioning

Observe section 6.1 *Safety instructions*.

Check the tightness of the whole system before startup.

6.2.1 System diagram

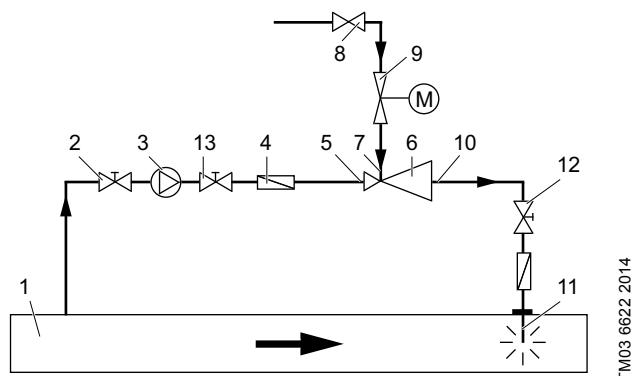


Fig. 5 System diagram

Pos.	Description
1	Supply line
2	Isolating valve
3	Motive water pump
4	Check-back valve
5	Motive water inlet injector
6	Injector
7	Gas supply
8	Safety pressure relief valve
9	Gas dosing system
10	Gas solution line (motive water outlet injector)
11	Injection unit
12	Isolating valve
13	Isolating valve

6.2.2 Checking the gas solution line for tightness

Observe section 6.1 *Safety instructions*.

This section refers to fig. 5.

- Loosen the union nut on gas supply (7).
- Disconnect gas supply (7).
- Open motive water isolating valves (2) and (13).
- Open isolating valve (12).
- Switch on motive water pump (3).
 - If no water escapes, the line is watertight.
- Reconnect gas supply (7).

If water escapes from between the motive water pump (3) and the injection unit (11), proceed as follows:

- Switch off motive water pump (3).
- Close motive water isolating valve (2).
- Close isolating valve (12).
- Repair the leak.
- Repeat this check.

6.2.3 Checking the diaphragm non-return valve

Observe section 6.1 *Safety instructions*.

This section refers to fig. 5.

- Loosen the union nut on gas supply (7).
- Disconnect gas supply (7).
- Open motive water isolating valves (2) and (13).
- Open isolating valve (12).
- Switch on motive water pump (3).
- Briefly shut isolating valve (12).
 - If no water escapes from the gas connection of injector (6), the diaphragm non-return valve is working properly.

Reconnect gas supply (7).

If water escapes from the gas connection of injector (6), the diaphragm non-return valve is faulty. Proceed as follows:

- Switch off motive water pump (3).
- Close motive water isolating valve (2).
- Repair the diaphragm non-return valve (see section 9.7 *Replacing spare parts*).
- Repeat this check.

6.2.4 Checking the operation of the injector

Observe section 6.1 *Safety instructions*.

This section refers to fig. 5.

- Loosen the union nut on gas supply (7).
- Disconnect gas supply (7).
- Open motive water isolating valves (2) and (13).
- Open isolating valve (12).
- Switch on motive water pump (3).
- Place your thumb over the gas connection of injector (6).
 - If suction can be clearly felt, the injector is working properly.

If injector (6) does not suck:

- Check the admission pressure and water quantity at motive water inlet (5).
- Check the counterpressure at gas solution line (10).
- Check the line lengths and cross-sections. See www.grundfos.com.
- See section 8. *Fault finding*.

If the error cannot be rectified, contact the Grundfos service.

7. Operation

Note

*The device is passive. All settings must be carried out on other components of the system.
Observe the manuals of the other components.*

8. Fault finding

Fault	Cause	Remedy
Maximum dosing flow is not achieved despite suitable size of injector.	Swirl piece in the nozzle is damaged or not present.	Insert new swirl piece.
	Counterpressure on the injector is too high (gas solution line is too long or line cross-section is too small).	Bring the injector closer to the injection unit, observe the line cross-sections. See www.grundfos.com .
	Motive water pressure or quantity is not set correctly.	Check the settings using the dimensioning tables. See section 11. <i>Dimensioning tables</i> .
	Dynamic pressure on the injector is too low due to line loss.	Check length and cross-section of motive water and gas solution line (see www.grundfos.com), use the least possible number of elbows.
	Process water temperature is higher than allowed.	Adjust process water temperature to a permitted value. See section 3.1 <i>General data</i> .
	Leakage in gas supply line.	Replace the gas supply line.

9. Maintenance

9.1 Safety instructions

Warning

Cleaning and maintenance must be carried out by authorised and qualified staff.

Shut down the whole system before any work at the system components and lines!

The system must be pressureless!

Make sure, that no residual gas is in the system.

Separate the whole system from the power supply before any work at the system components!

Wear the stipulated personal protective equipment (protective clothing, goggles, respirator etc.) when handling chemicals!

Observe the chemical manufacturer's safety data sheets (MSDS) and safety instructions of the used chemicals!

Make sure, that a sufficient amount of neutraliser (e.g. sodium thiosulphate for chlorine) is available on site.

Safety installations, which have been disabled during maintenance, must be enabled again immediately after maintenance.

Tightly close all gas-leading parts to prevent humidity from getting in.

Check the tightness of the whole system before recommissioning!



Danger of damage to the device!

Caution

Do not use sharp objects for cleaning.

9.2 Regular maintenance

This section refers to figs. 6 and 7. For injectors with counterpressure range 0 to 7 bar see fig. 6, for injectors with counterpressure range 7 to 10 bar see fig. 7.

Task	Interval	See section
Clean the device	If necessary	9.3
Visual inspection of injector and lines for leakage	Daily	
Check the tightness of the whole system	1 month	
Check and clean upstream dirt trap	3 month	
Check and clean the nozzle and the diffuser	3 month	9.4
Check the diaphragm non-return valve	3 month	9.5
Check the spring	3 month	9.6
Replace the spare parts	1 year	9.7

Additional cleaning and maintenance is required in case of malfunction.

Local circumstances (poor water quality, extreme pressure conditions) may shorten the maintenance intervals.

Note

9.3 Cleaning the device

If necessary, clean all product surfaces with a dry and clean cloth.

9.4 Checking and cleaning the nozzle and the diffuser

Observe section 9.1 *Safety instructions*.

Clean the removed parts thoroughly with warm water immediately after dismantling.

Caution

Before reassembling make sure, that all parts are clean, dry and undamaged!

Deposits prevent the injector from creating the required low pressure, which leads to a drop in output. The nozzle can be cleaned with hydrochloric acid. The following steps refer to fig. 5.

- Shut down the system.
 - Flush the system with clean water.
 - Close isolating valve (12).
 - Disconnect gas supply (7).
 - Loosen motive water connection (5) and gas solution line (10). On injectors with a hose connection, loosen the hoses.
 - Remove the injector from the system.
- The following steps refer to figs. 6 and 7. For injectors with counterpressure range 0 to 7 bar see fig. 6, for injectors with counterpressure range 7 to 10 bar see fig. 7.
- Remove union nuts (220) on both sides.
 - For injectors with counterpressure range 0 to 7 bar:
 - Remove nozzle (20). Leave swirl piece (25) in the nozzle.
 - For injectors with counterpressure range 7 to 10 bar:
 - Remove bushing (20) together with nozzle (21) and swirl piece (25). Leave swirl piece (25) in the bushing.
 - Clean the nozzle with hydrochloric acid.
 - Replace the nozzle, if it is worn.
 - Unscrew diffuser (10) and clean it with hydrochloric acid.
 - Replace the diffuser, if it is worn.
 - Reassemble the injector in the reverse order.
 - When reassembling, always use new gaskets and O-rings.

Reinstall the injector into the system and check the correct functioning. Observe sections 5. *Installation* and 6. *Commissioning*.

9.5 Checking the diaphragm non-return valve

Caution

Regular checks are necessary due to wear to the piston, seat and diaphragm.

To check the function of the diaphragm non-return valve, follow the instructions in section 6.2.3 *Checking the diaphragm non-return valve*.

9.6 Checking the spring

The spring is plastic-coated, which protects it against corrosion.

- Check the spring and make sure, that the plastic coating is not damaged.
 - For dismantling see the respective steps in section 9.7.1 *Dismantling*, for reassembling see the respective steps in section 9.7.2 *Reassembling*.

Caution

Replace the spring immediately, if the coating is damaged!

9.7 Replacing spare parts

Caution *Clean the removed parts thoroughly with warm water immediately after dismantling.*

This section describes the replacement of the parts included in the spare parts sets. For details on the spare parts sets, see section 10. *Spare parts*.

9.7.1 Dismantling

Observe section 9.1 *Safety instructions*.

The following steps refer to fig. 5.

1. Shut down the system.
2. Flush the system with clean water.
3. Close isolating valve (12).
4. Disconnect gas supply (7).
5. Loosen motive water connection (5) and gas solution line (10). On injectors with a hose connection, loosen the hoses.
6. Remove the injector from the system.

The following steps refer to figs. 6 and 7. For injectors with counterpressure range 0 to 7 bar see fig. 6, for injectors with counterpressure range 7 to 10 bar see fig. 7.

7. At injectors with counterpressure range 0 to 7 bar remove screw caps (200).
8. Remove screws (140).
9. Remove top part (90).
10. Remove diaphragm disc (170).
11. Remove spring (40).
12. Unscrew diaphragm ring (80) and remove diaphragm (130).
13. Unscrew screw ring (50).
14. Remove seat (60) including O-ring (180).
15. Unscrew piston (120) from top part (90).
16. Unscrew lower part (95) from body (30).
17. Unscrew diffuser (10) from body (30).
18. Unscrew union nuts (220) on both sides.
19. Only for injectors with counterpressure range 7 to 10 bar:
 - Remove bushing (20) including swirl piece (25). Leave the swirl piece in the bushing.
 - Remove nozzle (21).

9.7.2 Reassembling

Before reassembling make sure, that all parts are clean, dry and undamaged!

Caution *When reassembling, always use new gaskets and O-rings.*

Observe section 9.1 *Safety instructions*.

The following steps refer to figs. 6 and 7. For injectors with counterpressure range 0 to 7 bar see fig. 6, for injectors with counterpressure range 7 to 10 bar see fig. 7.

1. Only for injectors with counterpressure range 7 to 10 bar:
 - Replace O-ring (22) and insert nozzle (21) into body (30).
 - Insert bushing (20) including swirl piece (25) into body (30).
2. Replace O-rings (210) and fit union nuts (220) on both sides.
3. Replace flat gasket (240).
4. Grease the threads of the following connections:
 - Connection between the lower part (95) and the body (30).
 - Connection between the diffuser (10) and the body (30).

For drinking water applications, only food-safe grease must be used. The grease must be resistant against the chemicals used.

Caution

5. Replace O-ring (190).
6. Screw in body (30) into lower part (95) and tighten it carefully.
7. Screw in diffuser (10) into body (30) and tighten it carefully.

If the piston surface is damaged, the closing mechanism may not work.

Caution

Do not use tools when screwing in the piston.

8. Screw in new piston (120) by hand.
9. Replace O-ring (160).
10. Replace O-ring (70).
11. Insert new seat (60) with new O-ring (180).
12. Screw in new screw ring (50) and tighten it carefully with an appropriate face spanner.
13. Fit the new diaphragm (130) at diaphragm disc (170).
 - Make sure, that the orientation of the diaphragm is as shown in the respective figure. See fig. 6 or fig. 7.
 - Make sure, that the diaphragm fits into the diaphragm seat and is exactly centred on the diaphragm disc.
14. Screw in diaphragm ring (80) and tighten it carefully with an appropriate face spanner.
15. Carefully retighten screw ring (50).
16. Slide diaphragm disc (170) onto piston (120).
 - Diaphragm ring (80) must point to top part (90).
 - Turn diaphragm disc (170) so that the holes line up with the drill holes in top part (90).
17. Fit new spring (40).
 - Make sure, that the plastic coating of the spring is not damaged.
18. Fit top part (90).
19. Fit screws (140) and tighten them diagonally with a torque wrench.
 - Torque: 2.5 Nm
20. At injectors with counterpressure range 0 to 7 bar fit screw caps (200).

Reinstall the injector into the system and check the correct functioning. Observe sections 5. *Installation* and 6. *Commissioning*.

If water escapes: Tighten screw ring (50) a bit more.

10. Spare parts

Grundfos recommends to keep some spare parts on stock to make sure, that worn parts can be replaced quickly, when necessary.

10.1 Injectors for counterpressure range 0 to 7 bar

Description	Product number
Spare parts set for Cl ₂	96688748
Spare parts set for NH ₃ , SO ₂ , CO ₂	95701437

10.1.1 Sectional drawing

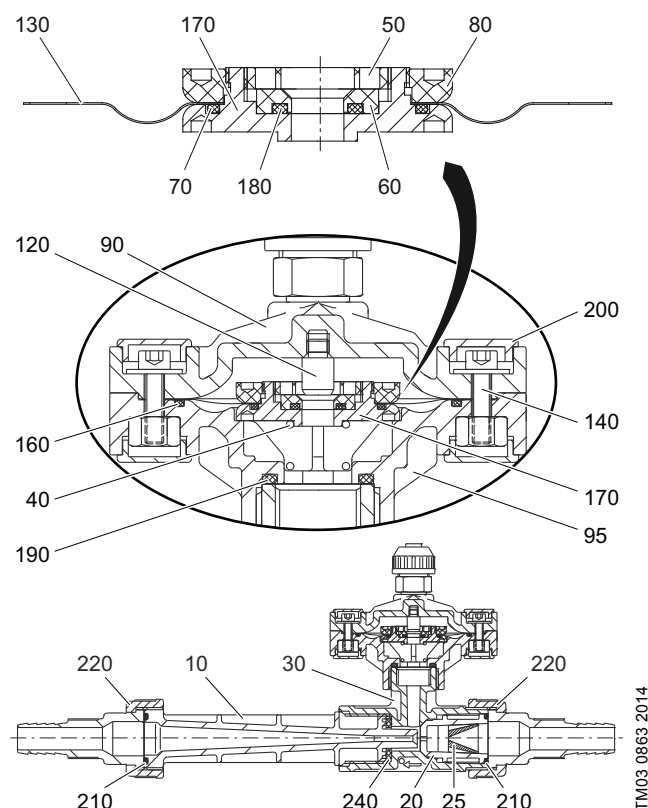


Fig. 6 Sectional drawing

Pos.	Designation	Pieces in spare parts set
10	Diffuser	not included
20	Nozzle	not included
25	Swirl piece (product no. 96681053)	not included
30	Body	not included
40	Spring	not included
50	Screw ring	1
60	Seat	1
70	O-ring	1
80	Diaphragm ring	not included
90	Top part	not included
95	Lower part	not included
120	Piston	1
130	Diaphragm	1
140	Screw	not included
160	O-ring	1
170	Diaphragm disc	not included
180	O-ring	1
190	O-ring	1
200	Screw cap	not included
210	O-ring	2
220	Union nut	not included
240	Flat gasket	1

10.2 Injectors for counterpressure range 7 to 10 bar

Description	Product number
Spare parts set for Cl ₂	95715145

10.2.1 Sectional drawing

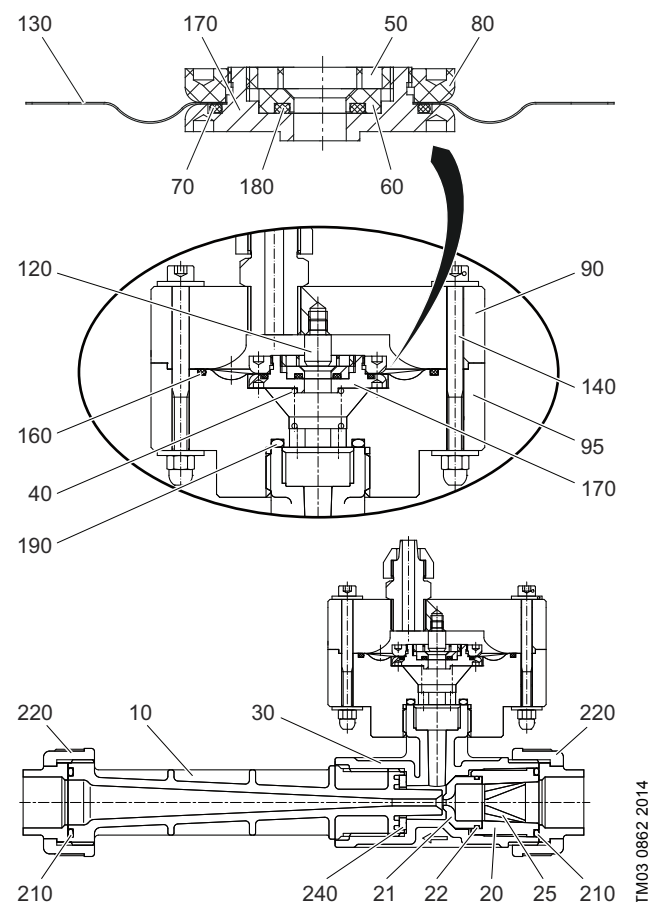


Fig. 7 Sectional drawing

Pos.	Designation	Pieces in spare parts set
10	Diffuser	not included
20	Bushing	not included
21	Nozzle	not included
22	O-ring	1
25	Swirl piece (product no. 96681053)	not included
30	Body	not included
40	Spring	not included
50	Screw ring	1
60	Seat	1
70	O-ring	1
80	Diaphragm ring	not included
90	Top part	not included
95	Lower part	not included
120	Piston	1
130	Diaphragm	1
140	Screw	not included
160	O-ring	1
170	Diaphragm disc	not included
180	O-ring	1
190	O-ring	1
210	O-ring	2
220	Union nut	not included
240	Flat gasket	1

11. Dimensioning tables

Criteria for the selection of an injector are (see fig. 4):

- Suction power at gas connection (1)
- Counterpressure at gas solution outlet (6)
- Motive water pressure at motive water inlet (3).

When dimensioning an injector, the required suction power and counterpressure is usually specified.

Other parameters such as motive water pressure and water quantity can be found in the following tables.

Example:

See section 11.1.1 *Injectors for counterpressure range 0 to 7 bar.*

To achieve a suction power of 500 g/h Cl₂ at a counterpressure of 2.5 bar and a motive water pressure of 7 bar, a water quantity of 530 l/h is required.

Note

All data refers to drinking water quality motive water.

11.1 Injectors for Vaccuperm systems WITH differential pressure regulator (VGA-113, VGA-117, VGS-141)

11.1.1 Injectors for counterpressure range 0 to 7 bar

Counterpressure [bar]	Motive water pressure [bar]	Water quantity [l/h] for suction power			
		up to 500 g/h Cl ₂ up to 500 g/h NH ₃ up to 500 g/h SO ₂ up to 210 g/h CO ₂	up to 1000 g/h Cl ₂ up to 1000 g/h NH ₃ up to 1000 g/h SO ₂ up to 350 g/h CO ₂	up to 2000 g/h Cl ₂ up to 1500 g/h NH ₃ up to 2000 g/h SO ₂ up to 550 g/h CO ₂	up to 4000 g/h Cl ₂ up to 2000 g/h NH ₃ up to 4000 g/h SO ₂ up to 800 g/h CO ₂
0	2	300	550	850	1200
0.5	3	360	600	1000	1450
1	4	410	670	1150	1600
1.5	5	450	750	1250	1850
2	6	490	820	1350	2000
2.5	7	530	880	1450	2150
3	8	560	920	1550	2300
3.5	9	590	1000	1650	2450
4	10	620	1050	1750	2600
4.5	11	650	1100	1850	2700
5	12	680	1150	1950	2800
5.5	13	710	1200	2000	2900
6	14	730	1250	2100	3000
6.5	15	760	1300	2150	3150
7	16	780	1350	2200	3300

11.1.2 Injectors for counterpressure range 7 to 10 bar

Counterpressure [bar]	Motive water pressure [bar]	Water quantity [l/h] for suction power		
		up to 500 g/h Cl ₂	up to 1000 g/h Cl ₂	up to 2000 g/h Cl ₂
5	11	950	1150	1700
6	12	980	1200	1750
7	13	1000	1250	1800
8	14	1050	1300	1900
9	15	1100	1350	1950
10	16	1150	1400	2000

11.2 Injectors for Vaccuperm systems WITHOUT differential pressure regulator (VGB-103, VGA-113, VGA-117)

11.2.1 Injectors for counterpressure range 0 to 7 bar

Counterpressure [bar]	Motive water pressure [bar]	Water quantity [l/h] for suction power		
		up to 500 g/h Cl ₂ up to 500 g/h NH ₃ up to 500 g/h SO ₂ up to 210 g/h CO ₂	up to 1000 g/h Cl ₂ up to 1000 g/h NH ₃ up to 1000 g/h SO ₂ up to 350 g/h CO ₂	up to 2000 g/h Cl ₂ up to 1500 g/h NH ₃ up to 2000 g/h SO ₂ up to 550 g/h CO ₂
0	2	550	850	1200
0.5	3	600	1000	1450
1	4	670	1150	1600
1.5	5	750	1250	1850
2	6	820	1350	2000
2.5	7	880	1450	2150
3	8	920	1550	2300
3.5	9	1000	1650	2450
4	10	1050	1750	2600
4.5	11	1100	1850	2700
5	12	1150	1950	2800
5.5	13	1200	2000	2900
6	14	1250	2100	3000
6.5	15	1300	2150	3150
7	16	1350	2200	3300

12. Additional non-return valve

In order to avoid a possible water entry into the gas supply line, an additional non-return valve can be mounted into the gas supply line. This permits to install the injector in any position. The additional non-return valve is installed as a syphon in the gas supply line.

Description	Product number
Complete non-return valve with connections	96727007
Non-return valve without connections	96617127

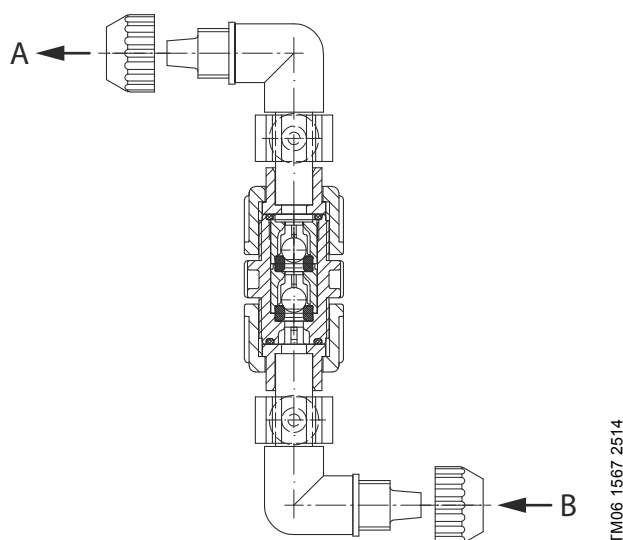


Fig. 8 Additional ball check valve

Pos.	Description
A	Line towards injector
B	Line from dosing regulator

13. Decommissioning

Caution

Seal up the inlet and outlet connections of the dismantled system components to make sure, that no moisture can ingress!

For details on decommissioning, see separate installation and operating instructions of the Vaccuperm system.

14. Disposal

This product or parts of it must be disposed of in an environmentally sound way. Use appropriate waste collection services. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

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